Syllabus for Practicum in Biostatistics (PHB 243b)

University of California, San Diego

School of Public Health
Winter Quarter 2026

Instructor Information

- **Instructor:** Ronald G. Thomas, PhD
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- **Office:** MTF 180b
- **Office Hours:** by appointment
- **Teaching Assistant:** Man Luo
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- **Course Time:** Tuesday and Thursday, 9:30–10:50 AM
- **Location:** Pepper Canyon Hall 245
- **Units:** 4
- **Prerequisites:** PHB 221 and PHB 243A

Course Description

This is the second course in a three-course sequence specifically designed for students enrolled in the MS in biostatistics program. Key goals are instruction on best practices in statistical collaboration and reproducible research.

Learning Outcomes

By the end of this course, students will be able to:

1. Develop and present detailed analysis plans and reports.
2. Apply statistical methods to real-world biomedical datasets.
3. Collaborate effectively in teams to produce reproducible research.
4. Communicate findings through professional-quality presentations.

Required Materials

All course materials will be posted on Canvas. Students will work with the following datasets:

Datasets

1. Penguin Dataset

- **GitHub Page:** https://github.com/rfordatascience/tidytuesday/blob/main/data/2020/2020-07-28/readme.md
- **Related:**

R Journal: Palmer Archipelago Penguins Data in the palmerpenguins R Package - An Alternative to Anderson's Irises https://journal.r-project.org/articles/RJ-2022-020/

2. ADNI Dataset

LONI archive site

ADNI | ADNI Data https://adni.loni.usc.edu/data-samples/adni-data/

Predicting conversion to Alzheimer's disease in individuals with Mild Cognitive Impairment using clinically transferable features | Scientific Reports https://www.nature.com/articles/s41598-022-18805-5

3. Stanford Open Policing Project

Data Page: https://openpolicing.stanford.edu/data/

Standardized data on vehicle and pedestrian stops from law enforcement departments across the United States.

Rivera R, Rosenbaum J. Racial disparities in police stops in US cities. *Significance*. 2020;17(4):4-5. https://rss.onlinelibrary.wiley.com/doi/full/10.1111/1740-9713.01412

Pierson E, Simoiu C, Overgoor J, et al. A large-scale analysis of racial disparities in police stops across the United States. *Nature Human Behaviour*. 2020;4(7):736-745. https://doi.org/10.1038/s41562-020-0858-1

4. Licorice Gargle

An R Package Of Medical Data For Teaching • medicaldata https://higgi13425.github.io/medicaldata/

A Randomized, Double-Blind Comparison of Licorice Versus Sugar-Water Gargle for Prevention of Postoperative Sore Throat and Postextubation Coughing

Anesthesia & Analgesia: https://pubmed.ncbi.nlm.nih.gov/23921656/

Required Software:

- R (latest version)

- Git and GitHub account

Class Schedule

Tuesday	Thursday
Jan 6 Introductions	**Jan 8** Introduction to Reproducible Research
Jan 13 renv and Docker	**Jan 15** Git and GitHub
Jan 20 Assign Project 1 (Penguins)	**Jan 22** Present P1 Plan
Jan 27 Test 1	**Jan 29** Present P1 Report
Feb 3 Assign Project 2 (ADNI)	**Feb 5** Present P2 Plan
Feb 10 Lecture and Lab	**Feb 12** Present P2 Report
Feb 17 Assign Project 3 (Policing)	**Feb 19** Present P3 Plan
Feb 24 Lecture and Lab	**Feb 26** Present P3 Report
Mar 3 Assign Project 4 (Licorice)	**Mar 5** Present P4 Plan
Mar 10 Test 2	**Mar 12** Present P4 Report

Assessment and Grading

Component	Percentage
Preparation and Presentation of Plans (4)	30%
Preparation and Presentation of Reports (4)	30%
Tests	30%
Class Attendance and Participation	10%

Presentations graded pass/fail. Lowest presentation grade will be dropped.

Grading Scale

- A+: 97-100%
- A: 93-96%
- A-: 90-92%
- B+: 87-89%
- B: 83-86%
- B-: 80-82%
- C+: 77-79%
- C: 73-76%
- C-: 70-72%
- D: 60-69%
- F: Below 60%

Course Structure and Expectations

This course consists of twice-weekly meetings combining lectures, discussions, and presentations. Students are expected to: - Attend all lectures and actively participate in discussions - Work collaboratively in assigned teams - Prepare and present analysis plans and reports - Provide constructive feedback on peer presentations - Complete all assigned readings before class

Course Policies

Attendance and Participation

Class attendance is mandatory and will directly impact your grade. Regular attendance and active participation are essential components of this course and account for 10% of your final grade. This includes engagement in discussions, team collaboration, and presentation feedback.

Each unexcused absence will result in a deduction from your participation grade. If you must miss class due to illness or emergency, please notify the instructor in advance when possible. Documented medical or family emergencies will be considered for excused absences on a case-by-case basis.

Presentation Guidelines

- Plans and reports will be presented during class time
- Each team will have 15-20 minutes for presentation plus 5-10 minutes for questions
- All team members must participate in the presentation
- Presentations should be professional, clear, and concise
- Slides should be submitted to Canvas prior to class

Electronic Devices

Laptops and tablets are welcome in class for note-taking and data analysis. However, please refrain from non-class activities that may distract you or others.

Communication

- Canvas will be used for announcements, assignments, and course materials
- Questions about course material should be posted on Canvas discussions so all students can benefit
- For personal matters, email the instructor directly
- Expect responses to emails within 48 hours during weekdays

Academic Integrity

All work submitted in this course must be your own and produced exclusively for this course. Academic dishonesty, including plagiarism, cheating, and unauthorized collaboration, will not be tolerated. While discussion of concepts and approaches is encouraged, sharing code for assignments is prohibited unless explicitly stated otherwise.

The use of sources (ideas, quotations, paraphrases, and code) must be properly acknowledged and documented. If in doubt, you are encouraged to review guidelines for the proper use of sources, as well as the UCSD policy on plagiarism and other forms of academic misconduct. It is your responsibility to know what constitutes academic misconduct at UCSD. The instructor is professionally and ethically responsible to report all possible integrity violations to the Academic Integrity Office.

UCSD's definition of academic misconduct can be found at: http://students.ucsd.edu/academics/academic-integrity/defining.html

The official university policy on the Integrity of Scholarship can be found at: http://students.ucsd.edu/_files/Academic-Integrity/Policy-on-Integrity-of-Scholarship_eff-fall2009.pdf

For more information, please refer to UCSD's Academic Integrity Office: https://academicintegrity.ucsd.edu/

AI Use Policy

Use of AI tools (such as ChatGPT, Claude, Copilot, etc.) is permitted in this course, provided that proper credit is given. When using AI assistance for analysis plans, reports, or code:

- Cite the AI tool used and the explicit prompts that were provided
- Clearly distinguish your original contribution from AI-generated content
- You remain responsible for the accuracy and quality of all submitted work
- AI-generated content should be reviewed, understood, and verified before submission

The goal is to learn how to use these tools effectively and transparently as part of modern biostatistical practice, not to substitute them for your own understanding and critical thinking.

Accommodation for Students with Disabilities

UC San Diego complies with the Americans with Disabilities Act and Section 504 of the Rehabilitation Act. If you have a documented disability for which you require accommodation, please contact the Office for Students with Disabilities (OSD) as soon as possible: https://osd.ucsd.edu/

After registering with OSD, please discuss your specific needs with the instructor early in the quarter to ensure appropriate accommodations can be made.

Diversity and Inclusion

Diversity enriches our academic community and is essential for excellence in research and education. We are committed to fostering an inclusive classroom environment that values the diverse backgrounds and perspectives of all students. We expect all class participants to contribute to a respectful, welcoming, and inclusive environment.

Basic Needs and Wellness Resources

UCSD recognizes that students may experience challenges related to food insecurity, housing, health, or other basic needs that may interfere with academic performance. If you find yourself facing such challenges, please consider the following resources: - Basic Needs Hub: https://basicneeds.ucsd.edu/ - Counseling and Psychological Services (CAPS): https://wellness.ucsd.edu/caps/ - Student Health Services: https://wellness.ucsd.edu/studenthealth/

Technology Support

For technical assistance with course technology, contact the UCSD IT Service Desk: - Email: servicedesk@ucsd.edu - Phone: 858-246-4357 - Website: https://support.ucsd.edu

UCSD Policies and Resources

1. **UCSD Principles of Community:** Link to UCSD Principles of Community
2. **Policy on Integrity of Scholarship:** Link to Policy
3. **Counseling and Psychological Services (CAPS):** Link to CAPS
4. **Office for Students with Disabilities (OSD):** Link to OSD

Disclaimer

This syllabus is subject to change. Any modifications will be announced in class and on Canvas. It is the student's responsibility to stay informed about changes to the schedule or policies.